UNIVERSITY of York

School of Physics, Engineering and Technology

ELE00050I Music Technology Project 2 Assessment 2024/25

Summary Details

This assessment **Music Technology Project 2** contributes **100%** of the assessment for this module. However, it is split into 4 components as follows, all of which are described in this single document:

- Initial Report (Group, 10%)
- Tender Presentation Video (Group, 10%)
- Final Report (Individual, 60%)
- Demonstration Video (Group, 20%).

Clearly indicate your **Name** on every separate piece of work submitted.

Submission is via the VLE. **The components deadlines are shown on the Assessment Calendar.** Please try and submit early as any late submissions will be penalised.

Academic Integrity

It is your responsibility to ensure that you understand and comply with the University's policy on academic integrity. If this is your first year of study at the University then you also need to complete the mandatory Academic Integrity Tutorial. Further information is available at http://www.york.ac.uk/integrity/.

In particular please note:

- Unless the coursework specifies a group submission, you should assume that all submissions are individual and should therefore be your own work.
- All assessment submissions are subject to the Department's policy on plagiarism and, wherever possible, will be checked by the Department using Turnitin software.

ELE00050I

MUSIC TECHNOLOGY PROJECT 2 - 2024-2025

AUDIO SYNTHESIS ASSIGNMENT

1. Assignment Structure

This assignment (Coursework) contributes 100% of the assessment for this module, and consists of four components, spread throughout the module as follows:

- Week 5: Initial Report (Group, 10%)
- Week 7: Tender Presentation Video (Group, 10%)
- Week 13: Final Report (Individual, 60%)
- Week 15: Demonstration Video (Group, 20%).

Please note that the submissions are *not* anonymous, so you should use your name and the names of your team members. Also note that for the group components, the same mark will be given to all group members provided they make some contribution to the submission. (If any group member makes no contribution to a group component, that student's mark will be zero for that component.)

2. Assignment Scenario

The scenario for this assignment is that you are a start-up company bidding to win one of the UK Government's contracts to supply hardware, software, and educational materials for their new *Electronics in Schools* initiative. You are bidding for the Music Technology (Audio Synthesis) subsection, which uses simple electronic components in conjunction with a low-cost microcontroller to provide a working system for audio synthesis and control.

The Call to Tender document is provided here as an Appendix. You should read this thoroughly to understand the scenario, but your actual assignment requirements are outlined in detail below.

Note that a formal document responding to the Call to Tender is not required for this project. In terms of directly responding to the tender, only the Tender Presentation Video is required.

3. The Impact of Engineering on Society

In the rapidly evolving field of engineering, it is important to address not only the technical aspects but also the societal implications of your innovations. This project has been designed to integrate the five key societal engineering themes that have been defined by the UK Engineering Council:

1. Sustainability,
2. Ethics,
3. Equality/Inclusion,
4. Risk, and
5. Security

(which we sometimes abbreviate as SEERS).

These themes help us all to develop an understanding of how our engineering solutions affect the world around us. By considering **Sustainability**, we ensure that our projects are environmentally responsible. **Ethics** helps us navigate the moral implications of our technological choices. **Equality** and **inclusion** ensure our designs are accessible to everyone, irrespective of their abilities or backgrounds. Predicting **Risk** prepares us to anticipate potential failures or bad outcomes, and prepare solutions. Lastly, focusing on **Security** ensures the integrity and safety of our technological solutions.

Assessing these aspects in your projects broadens your learning experience and prepares you to become thoughtful, responsible engineers who can contribute positively to society. We encourage you to integrate these themes into your projects, and they will form a key part of your assessment criteria.

We ask you to consider these issues as a Group, and to report on them in your Initial Group Report (see Section 5.1 below) by downloading, filling in, and including the Table that we have drafted here. In addition we ask you to reflect Individually on each of these issues in the final report.

4. Assignment Tasks

Your task is to design, develop and test a microprocessor-based audio system which:
- Has physical sensors which allow a user to control sound generation and processing in real-time;
- Creates sound by running audio generation algorithms (synthesis or sample playback) on the STM32F4 microprocessor.

You should choose one of the following application areas for your system:
- A *musical instrument* (the user operates physical sensors which control the parameters of your onboard synthesis system with the aim of creating music performances in real time)
- An *interactive sonification explorer* (the user operates physical sensors which control the way in which pre-loaded data is converted to sound).

The Government have indicated that price is a very important factor for them, and you should therefore design your product to minimise the cost of manufacture. They have also identified that a much wider market exists in the developing world for low-cost educational equipment and are keen to replicate the success of the BBC micro:bit. It is therefore very important that you prioritise this factor in your design.

You are to work in a group (which will be formed at the start of Semester 1) following the principles covered in the associated project management lectures. The aim of the group work is to give you practical experience of working in teams on a complex task, where each of you are assigned specialist roles or areas of responsibility. Part of the assignment involves an element of peer assessment, which is outlined later in this document. A good group will work together consistently throughout the project period, will have good procedures and management in place, and will allocate and monitor individuals' workload and contribution throughout the process.

Even though you are asked to submit an individual report at the end of the project to demonstrate your understanding and progress, you should NOT think of this assignment as several individual projects – but rather as *one coordinated activity*, spread out between multiple people, using common code and equipment. This gives you good experience for the ways in which companies work.

4.1 Technical Research

This assignment scenario has been set up to emulate the conditions that you might find in a company where you are given a complex technical task to carry out, but you do not currently have training in all of the required technical areas. In order for you to carry out this task effectively, you will need to:
- Find out about how digital synthesis and sampling can be implemented on a microprocessor.
- Read up on how the STM32 microcontrollers operate for audio processing, generation and output.
- Investigate how to design and build either a computer-based musical instrument, or a sonification exploration device.

This will give you insight into how professional music technology companies produce digital synthesisers.

You may wish to take this further than synthesis and sampling if you have time / interest. For instance, you could look into how digital filters are made, and implement one or more for your system. Similarly, you could investigate digital audio effects such as reverb, chorus, delay, flanging etc.

In deciding on your specification, you should aim to strike a balance between:
- Engineering ambition and challenge.
- Ensuring that you can get something working for the demonstration.
- Producing an exciting product that will enthuse and inspire the next generation of music technology and audio engineers.

4.2 Product Research, Design and Development

You also need to carefully consider the educational scenario in which this assignment is set.

4.2.1 Project Overview:

In response to the UK Government's initiative to include Electronics as part of the standard curriculum in secondary schools from 2025 onwards[1], your assignment is to develop an innovative and engaging sound generation device for use in the audio synthesis subsection of the curriculum. The project will involve coding a microprocessor for sound generation, building an electronic interface for sound control, and designing a user-friendly, interactive device suitable for educational purposes. Your team has the option to create either an electronic musical instrument or a sonification device for interactively exploring data as sound.

4.2.2 Project Objectives:

a) Design an innovative electronic musical instrument or sonification device suitable for secondary school Electronics education.
b) Develop and test a functional prototype of the chosen device, integrating microprocessor-based sound generation and analogue electronic control interfaces.
c) Create engaging instructional materials to support educators in using the device effectively in the classroom.
d) Prepare a Tender proposal for the audio synthesis component to be submitted to the UK Government, in the form of a short video.
e) Document your work and evaluate and reflect on your team operation and product context in both a Group Demonstration Video and an Individual Report.

4.3 Supervision

Your group should organise weekly meetings during which each team member reports on the progress they have made that week, and tasks for the next week are decided. Any problems and concerns that have arisen during the course of the week should be discussed and actions assigned. The minutes of these meetings should be recorded.

A template for these minutes is available on the module VLE site. You will receive feedback on these documents from your allocated project mentor. These documents can be used as source material for your final individual reports (and they should be included as an appendix in these reports).

Your group is also required to complete a shared worksheet at the end of every week of the project, detailing the progress made that week by each team member, their plans for the next week, and any problems or concerns that have arisen. (A template for these reports is also available on the module VLE site.) These documents can also be used as source material for your final individual reports.

You are encouraged to contact the Module Coordinator (andy.hunt@york.ac.uk) to set up a group meeting to discuss your project ideas. Three of these are to be held, one in each of Weeks 2, 5 and 8.

[1] Sadly fictional.

5. Assignment Deliverables

This section outlines the requirements for the four components of the assignment, in the order of their deadlines.

On each subsequent page are the detailed descriptions for:

- Initial Report (section 5.1)
- Tender Presentation Video (section 5.2)
- Final Report (section 5.3)
- Demonstration Video (section 5.4)

5.1 Initial Report (Group submission, 10%, due in Week 5)

The Initial Report is a **group** report which describes your team and your initial planning for the project. You have a word limit of 2500 words in total (not including Title Page, Contents or References), but you are encouraged to use as many diagrams as you like (plans, company structure charts, ideas sketches, Gantt chart, tables etc).

The structure and layout of the report is entirely up to you, but should cover the following issues (which also form the Marking Scheme for this report, all equally weighted):

1. **Your Team:** Group structure, company name, logo etc.

2. **Educational Application:** Describe your initial ideas on how you plan to meet the Government scenario (see Appendix 1). Develop a concept for the chosen device including a list of features and design considerations tailored for educational purposes.

3. **Product Specification:** What will your product do? What components and parts (hardware and software) will it have, and how will it operate technically?

4. **Costs:** What is the target cost for manufacture of your product? Provide a breakdown of the costs of any components used. Do a simple estimation of production costs.

5. **Project Management:** Describe how your project will be organised and managed. You should consider providing a) a table of task allocation between students / roles b) a Gantt chart – with overall timing and milestones, and c) a risk analysis.

6. **Societal Impact:** Download this Table, and investigate each category as a team. Fill in the table with your conclusions for your product and team, and include the edited table in this report. For **Ethics**: summarise all ethical considerations for your product, and its use in schools. For **Environmental Considerations**: Describe the impact of your product on the environment. Consider the product's entire lifecycle from sourcing components, manufacturing, active operation, and end-of-life (re-use, recycle, disposal). Refer to the main official directives on this subject and demonstrate that you understand the effect of your product on the environment and have sought to minimise any negative effects.

Remember that in a Group Report everyone should be contributing equally. There are many roles which can be assigned to various team members including:

- Ideas brainstorming
- Time-planning and coordination
- Analysing the project Scenario
- Describing your approach
- Writing technical plans
- Producing diagrams
- Re-drafting and compression to meet word-count limit.
- Proofreading, checking and editing.

Just one person from each group (selected by the team) should submit the Group Initial Report – in PDF format – to the submission point on the VLE.

5.2 Tender Presentation Video (Group submission, 10%, due in Week 7)

Your initial response to the tender submission takes the form of a video presentation that communicates your proposal and demonstrates your team's capabilities. The video should be engaging, professional, and visually appealing. The customer (the Government) will be reviewing these initial submissions to decide on a shortlist of bidders that they will take forward to the next stage of the tender process.

Below are guidelines on how to approach this task and the elements that should be included to make the best impression (and which also form the Marking Scheme for this report, all equally weighted):

a) **Introduction:**
   - Begin the video by introducing your start-up company, including its mission, vision, and any relevant experience in developing electronic musical instruments or sonification devices[2].
   - Introduce your team members, mentioning their roles, qualifications, and expertise.

b) **Concept Presentation:**
   - Clearly present the target specification of your proposed instrument or sonification device.
   - Explain the unique features, specifications, and design considerations tailored for educational purposes.
   - Use visuals, such as concept sketches, 3D renderings, or animations to enhance the presentation.

c) **Development Plan:**
   - Outline your proposed project plan, including timelines, milestones, and the development process for the design, prototyping, and testing phases.
   - Emphasise your team's technical expertise and the strategies you will employ to ensure the project's success.

d) **Educational Materials:**
   - Provide an overview of the educational materials you plan to develop, possibly including lesson plans, user manuals, and teacher training materials.
   - Explain how these materials could help with the Music Technology (Audio Synthesis) subsection of the secondary school Electronics curriculum.

e) **Conclusion:**
   - Summarise the main points of your proposal, explaining the benefits of your innovative device and your team's expertise.
   - Provide a clear call to action, encouraging the Government to choose you for the contract.
   - At the end of the video, display your company's contact information, including your email addresses (and any company links such as a website).

f) **Production Quality, Length & Format:**
   - Ensure the video is well-produced, with clear audio, steady camera work, and careful editing.
   - Use on-screen text, graphics, or animations to emphasise key points.
   - The video should be a maximum of 20 minutes, but nearer 10 minutes is probably ideal.
   - It should be in a widely accepted format (e.g., mp4) with high resolution (at least 720p).

Keep in mind that the video presentation is an opportunity to showcase your start-up's creativity and professionalism. Be sure to rehearse and refine the presentation to ensure it is polished, engaging, and communicates your proposal effectively. Pay attention to details such as lighting, sound quality, and on-screen visuals to make a lasting impression on the evaluation committee.

By following these guidelines and presenting a well-structured, visually appealing video, you would increase your chances of winning the tender for this innovative project in the Electronics in Schools Initiative (and raise your marks in this educational scenario).

---

[2] This could, for example, include your first-year project, or any devices you have built with ShockSoc.

5.3 Final Report (Individual submission, 60%, due in Week 13)

This report must be written individually and should detail your individual contribution to the project, as well as describing the group work as a whole.

5.3.1 Report Length & Purpose

Your report should be *between 3000 and 5000 words* in length. This word-count does ***not*** include the title page, table of contents, references or any appendices, but does include figure captions and section titles. Even though you will have been working on a good deal of technical material it is always important as engineers to be able to summarise a whole project in a concise report. This report assignment gives you practice and feedback on that skill. You may include Appendices (e.g. for full code listings, or further information about educational resources) but do not use Appendices for core information that should be in the report as specified here. More details of the required contents are listed in the marking scheme below.

5.3.2 Report Format & Organisation

Produce a PDF file for your report, using A4 page size with 1.5 cm margins all round, 11-point text single line spacing, with paragraphs justified to both margins. Begin with a front page with the school, project title, date, group number, members, and your name. You should include a table of contents page.

5.3.3 Report Content & Marking Scheme

Write an individual report which covers the following sections (shown here with their relative weightings):

a) Product Concept (10%) (main aim, who it's targeted at, marketing ideas)

b) Description of Entire System (10%) (technical overview, specification, use of audio synthesis**)**

c) Analysis of Effectiveness & Future Plans (10%) (achievements of the prototype compared to the specification in the initial report, quality of sound and user interfacing, further work needed as a group to complete the product prototype)

d) Individual Contribution (40%)

   a. **Clear statement of individual role and List of contributions** to the project. This should include references to your progress statements in the weekly minutes and worksheets. Please also include a list of the main roles taken by the other team members.

   b. **Technical detail of individual contributions**
   Here, describe the work that you have personally contributed (or jointly worked on, making clear which is your own work). If you have produced code and/or configured hardware, then include code, photos and test data. If you have not made as much progress as initially expected, then describe your *progress, and your plans for future work* in enough technical detail that someone else could complete the work in a well-equipped laboratory. For this, you should contemplate including some of the following as appropriate: Pseudocode and flowcharts / Written (but as yet untested) code / Test plans / Circuit diagrams and simulations /CAD designs for enclosures or circuit boards etc).

e) Reflective Analysis and Peer Review (20%) – see section 6 below.

f) Overall Appearance and Structure (10%)
Try to produce something that you would be comfortable passing on to your boss, after you have been asked for a report on a recent project. Do some research on different possible layouts for your report. Use plenty of diagrams. Get some feedback on your report’s readability and format from your supervisor, but you can also ask fellow students.

5.4 Demonstration Video (Group submission, 20%, due in Week 15)

As part of your final assessment, you are required to produce a group demonstration video showcasing your project's progress and achievements. Think of this as an academic demonstration and reflection video. It is not intended to be shown externally (to the Government, for instance as in the Tender Video). This video should briefly explain your product, demonstrate its functionality, and highlight any challenges or accomplishments. The following guidelines will help you create a well-organised and informative video presentation:

a) **Video Length and Format, and Production Quality:**
   - The video should be between five and ten minutes in length.
   - Ensure the video is in a widely accepted format (e.g., mp4) with a resolution ≥ 720p.
   - While we understand that you may be using mobile phones for filming and free software for editing, aim for the best audio and video quality possible within these limitations.
   - Use on-screen text, graphics, or animations to emphasise key points and enhance the overall presentation.

b) **Introduction:**
   - Begin the video by introducing your team and providing a brief overview of the electronic musical instrument or sonification device you have developed.
   - Mention the main functions and any unique features or specifications of your product.

c) **Product Demonstration:**
   - Demonstrate the functionality of your product, including any sections that may only partially work.
   - Clearly explain how the different components of your device interact and contribute to the overall system.
   - Show how your product can be used in an educational setting or integrated into the Electronics curriculum (particularly the Audio Synthesis subsection).

d) **Achievements and Challenges:**
   - Highlight any significant accomplishments or milestones achieved during the project.
   - Discuss any challenges faced and the solutions or strategies employed to overcome them.

e) **Group Involvement:**
   - Ensure that all group members contribute to the video presentation, whether through on-screen appearances, voiceovers, or behind-the-scenes roles such as script-writing, story-boarding, editing or filming.
   - Showcase the collaborative effort and teamwork involved in the project's development.

f) **Conclusion:**
   - Summarise the main points of your demonstration, emphasising the achievements and potential impact of your product in the Electronics in Schools Initiative.
   - Express gratitude for the opportunity to participate in the project and your team's enthusiasm for the future development or improvement of the product. This is a respectful part of any tender presentation.

By following these guidelines and presenting a well-organised, informative demonstration video, you will effectively showcase your team's efforts and accomplishments.

Aim for clarity and coherence in your video to make a positive impression and achieve the best possible outcome.

6. Reflective Analysis and Peer Review

In your Individual Report, you should include a substantial section where you analyse and reflect on how well:

1. your group was structured (Group Roles)
2. it worked in practice (Project Management)
3. you and your colleagues contributed individually (Peer Review)
4. your product was oriented to the societal issues (SEERS)
5. your own work process has developed (Strengths)

6.1 Reflection on Group Roles

Reflect on how well your group structure was suited to the project. What worked and what didn't? If you could reorganise the group based on what you know now, what would you do differently? Consider the results of the Belbin group roles assessment you completed at the beginning of the project. Did team members actually take on the roles they were assessed for, or did different roles emerge as the project progressed? How did this impact the overall effectiveness and collaboration within your group?

6.2 Project Management

Reflect on your time management and project planning, especially in the light of any unforeseen circumstances that have occurred since you started the project. How has your group coped with the change, shown resilience, and reconfigured its working operations?

Also include any Gantt chart or Work Breakdown Structure that was produced at the start of the project, and comment on your progress against it. Comment on how well your Risk Register helped you mitigate any problems that did occur, and anticipate and avoid any problems that might otherwise have occurred.

6.3 Peer Review

Give a mark to ALL students in the group, including yourself, as if you had 100 shares to distribute. Allocate shares according to your perception of the effort, effectiveness and overall contribution to the group that each individual made over the lifetime of the project. For example, if you believe that the 5 students in your group worked equally well, then allocate them 20 shares each. However, a distribution of *40, 20, 20, 20, 0* implies that the first student contributed twice as much as the next three, and the final student contributed nothing.

Add a short paragraph for each student justifying your share allocation. Note: these reviews and share allocations are private and will not be fed back to the other group members. They will be used to inform the marking panel about the perceived effectiveness of individual contributions to the group.

6.4 Societal Impact (SEERS)

Revisit Section 3 of this assessment document. Write a paragraph for each of the 5 societal issues (SEERS) evaluating the key issues raised in each during your project work, and highlighting any further work that you think needs to be done in these areas.

6.5 Reflection on Individual Strengths

Think back to your York Strengths evaluation in last year's project. Reflect on how you have used your strengths and how they have developed from year 1. Describe the impact of using your strengths in your project/group work/other activities, in particular considering how you would communicate these to an employer (e.g. in an interview, asking how you developed in a recent project). Reflect on how your strengths would translate into different environments and the world of work.

7. Submission

Your **Reports** - in PDF format - should be uploaded to the VLE by the date shown in the statement of assessment.

- For the Initial Report (Group) just one person from each group (selected by the team) should submit the report – in PDF format – to the submission point on the VLE.
- For the Final Report (Individual) each of you should submit your own report (again in PDF format) to the submission point.

Your **Group Videos** should be uploaded to the special VLE submission point for which you will be emailed a link closer to the deadline. Just one person from each group (selected by the team) should submit the video.

APPENDIX 1

Call to Tender

Development of an Innovative Electronic Musical Instrument or Sonification Device for the Audio Synthesis Subsection of the Electronics in Schools Initiative

Ref: UKG/MT/2024/07

1. Introduction

The UK Government, as part of its commitment to enhance the educational experience of secondary school students, is launching the *Electronics in Schools* Initiative from 2025 onwards. As part of this initiative, the Government invites start-up companies to submit tender proposals for the development of a low-cost yet innovative electronic musical instrument or sonification device tailored for the audio synthesis subsection of the Electronics curriculum. The selected contractor will be responsible for delivering detailed design specifications, a working prototype, and accompanying educational materials.

2. Scope of Work

The successful tenderer will be required to:

2.1. Design an innovative electronic musical instrument or sonification device suitable for secondary school Electronics education.

2.2. Develop and test a functional prototype of the chosen device, integrating microprocessor-based sound generation and analogue electronic control interfaces.

2.3. Create engaging instructional materials to support educators in using the device effectively in the classroom.

2.4. Explore the potential applications of the chosen device within the broader context of music technology and engineering education worldwide.

3. Deliverables

If successful, the chosen contractor will be required to provide the following deliverables:

3.1. A detailed design specification document for the chosen device, including specifications, features, and design considerations tailored for educational purposes.

3.2. A fully functional prototype of the chosen device, demonstrating the integration of microprocessor-based sound generation and analogue electronic control interfaces, together with fully documented source code of any software required to use the device.

3.3. A comprehensive user manual with clear instructions for operating the chosen device.

3.4. Lesson plans and educational resources that effectively incorporate the use of the device in classroom activities.

3.5. Teacher training materials to facilitate the integration of the device into classroom instruction.

3.6. A report outlining potential applications of the chosen device within the broader context of music technology and education, along with recommendations for future improvements or modifications based on feedback from educators and students.

4. Tender Process and Submission Requirements

Organisations interested in responding to this Call to Tender must initially submit a short video (approximately ten minutes long) describing their proposed approach and their relevant capabilities and experience. After reviewing these submissions, the Government will select a shortlist of companies to take forward to the next stage of the process.

Organisations should prepare a tender response which contains the following information:

4.1. Company profile, including experience in the development of electronic musical instruments or sonification devices, as well as any relevant expertise in the field of education.

4.2. A conceptual design of the chosen device, including a description of its unique features and detailed specifications.

4.3. A proposed project plan, including timelines and milestones for the design, development, testing, and delivery of the prototype and accompanying educational materials.

4.4. A pricing proposal, including a detailed breakdown of costs associated with the design, development, and delivery of the chosen device and accompanying materials.

4.5. A description of the proposed team, including the roles, qualifications, and experience of each team member.

The Government will shortlist companies for the final contract, based on the success of this initial tender video.

5. Evaluation Criteria

Tender proposals will be evaluated based on the following criteria:

5.1. Innovation and suitability of the proposed device for secondary school Electronics and Music Technology education.

5.2. Technical expertise and experience of the start-up company in developing electronic musical instruments or sonification devices.

5.3. Quality and comprehensiveness of the proposed project plan and pricing details.

5.4. Suitability of the proposed educational materials for the secondary school Electronics curriculum.

5.5. Potential for the use of the chosen device outside the needs of schools in the UK.

6. Terms and Conditions

The UK Government reserves the right to accept or reject any or all tender proposals, to negotiate with any or all tenderers, and to award the contract in the best interests of the *Electronics in Schools* Initiative.

The UK Government also reserves the right to cancel or amend this Call to Tender without any liability to tenderers.